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Training load monitoring in team sports: a practical approach to addressing missing data

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ABSTRACT

Training load (TL) is a modifiable risk factor that may provide practitioners with opportunities to mitigate injury risk and increase sports performance. A regular problem encountered by practitioners, however, is the issue of missing TL data. The purpose of this study was to examine the impact of missing TL data in team sports and to offer a practical and effective method of missing value imputation (MVI) to address this. Session rating of perceived exertion (sRPE) data from 10 male professional soccer players (age, 24.8 ± 5.0 years; height, 181.2 ± 5.1 cm; mass, 78.7 ± 6.4 kg) were collected over a 32-week season. Data were randomly removed at a range of 5–50% in increments of 5% and data were imputed using 12 MVI methods. Performance was measured using the normalized root-mean-square error and mean of absolute deviations. The best-fitting MVI method across all levels of missingness was Daily Team Mean (DTMean). Not addressing missing sRPE data may lead to more inaccurate calculations of other TL metrics (e.g., acute chronic workload ratio, training monotony, training strain). The DTMean MVI method may provide practitioners with a practical and effective approach to addressing the negative consequences of missing TL data.

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KEYWORDS
Training load; missing data; team sports; injury; performance

Introduction

Injury has known detrimental effects on player welfare (Gouttebarge et al., 2016; Manley et al., 2017; McCrory et al., 2017; Rice et al., 2016) and team and individual athletic performance (Drew et al., 2017). Subsequently, there has been a justifiable increase in research exploring methods of minimizing injury occurrence (Drew & Finch, 2016; Eckard et al., 2018; McCall et al., 2018). The multifactorial complex nature of sports injuries is derived from both modifiable (e.g., strength, flexibility) and non-modifiable (e.g., age, sex) inherent risk factors (Bahr & Krosshaug, 2005; Meeuwisse, 1994). One such modifiable risk factor is training load (TL), which can be defined as the cumulative amount of stress placed on an athlete from single or multiple training sessions over a period of time (Eckard et al., 2018; Soligard et al., 2016).

The existence of a relationship between TL and injury is firmly established (Drew & Finch, 2016; Eckard et al., 2018). Training is essential for preparing players for the demands of the sport and TL that inadequately prepares players for the demand of match play could place them at a greater risk of injury and hinder performance (Ball et al., 2018). However, a sudden increase or "spike" in TL may be associated with an increase in incidences and severity of injury (Drew & Finch, 2016; Griffin et al., 2020a). As a result, TL can be an indicator of injury and therefore provide strength and conditioning coaches and sport scientists with possible opportunities to reduce the risk of injury. Justifiably, practitioners working across individual and team sports include TL in their player monitoring practices (Akenhead & Nassis, 2016; Comyns & Hannon, 2018; Griffin et al., 2020b; Taylor et al., 2012). Despite being good practice, monitoring TL brings several challenges (Griffin et al., 2020b), an example of one such challenge is that of missing data (Jaspers et al., 2018; Windt et al., 2018).

The optimal method of monitoring TL is not yet fully understood but the International Olympic Committee consensus statement on load in sport and risk of injury recommends the use of a TL monitoring system that allows the best possible configuration of training, competition and other stressors external to sport in order to maximize adaptation and performance while minimizing the risk of injury (Soligard et al., 2016). Monitoring should also be performed consistently to allow adjustment to TL prescription, but without overburdening the players (Soligard et al., 2016).

TL can be categorized as either internal or external. Internal TL is the physiological and/or psychological stress imposed on the athlete during training and/or competition (e.g., session rate of perceived exertion (sRPE), heart rate, blood lactate), whilst external TL is the objectively measured work performed independent of internal workloads (e.g., speed, acceleration, distance) (Soligard et al., 2016). sRPE is the product of training intensity, measured using Borg's (1982) 0–10 scale and duration in minutes (Foster et al., 2001), and provides a measure of TL in arbitrary units (AU). It is a common measure of TL used by practitioners working with teams (Comyns & Hannon, 2018; Griffin et al., 2020b; Soligard et al., 2016) as it is inexpensive

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and non-invasive while broadly supported as a valid and reliable means of calculating TL (Herman et al., 2006). Additional metrics can be derived using sRPE data including weekly TL, weekly changes in TL, acute:chronic workload ratio (ACWR), training monotony and strain. Importantly, subjective measures have been shown to be more sensitive and consistent in determining acute and chronic changes in a player's wellbeing in response to loads when compared to objective measures (e.g., heart rate) (Saw et al., 2016). Additionally, a recent systematic review of the relationship between TL and injury found 21 studies that examined sRPE and injury reported statistically significant results, leading to the conclusion that sRPE provided the strongest evidence of all TL measures examined (Eckard et al., 2018).

A considerable problem that practitioners regularly encounter is the issue of missing TL data (Windt et al., 2018). Achieving complete adherence from players can be difficult, with Windt et al. (2018) suggesting missing TL data is a near certainty when collected in applied settings. Missing data is particularly common when data collection involves the use of self-report measures (Collins, 2006; Fox-Wasylyshyn & El-Masri, 2005). For example, when recording sRPE for every individual within a team, players will occasionally fail/forget to record their TL data after a training session or match. These missing data and resulting incomplete dataset may alter the practitioner's understanding of a player's response to training and in turn, the accuracy of training strategies (Weston et al., 2015). Without a complete dataset, all other metrics (e.g., weekly changes in TL, ACWR, etc.) calculated using these sRPE data would be inaccurate. This also presents considerable challenges for the practitioner in terms of individualized TL prescription.

To address missing data, multiple imputation methods have been suggested whereby missing values are replaced with respective estimates that are based on the values of other variables/items within the dataset (Collins, 2006; Fox-Wasylyshyn & El-Masri, 2005). This is known as missing value imputation (MVI) (Lin & Tsai, 2020). Data imputation is based in statistical theory and, if the missing data treatment is independent of the algorithm used, it can eliminate potential bias (Collin, 2006; Batista & Monard, 2003). This is particularly true if variables that are strongly correlated with those missing are included in the analysis (Collins, 2006). Several methods of MVI have been examined, namely; sample and group mean substitution, case mean substitution, hot deck, cold deck, last observation carried forward (LOCF), regression imputation, maximum likelihood (ML), expectation maximization (EM) and multiple imputation (Batista & Monard, 2003; Lin & Tsai, 2020). Each of these methods have respective strengths and limitations (Newgard & Lewis, 2015). For instance, while simple imputation methods such as LOCF and sample and group mean substitution are less complicated to compute, they may fail to account for the uncertainty in imputing missing values and can cause spurious results (Newgard & Lewis, 2015).

Missing data are typically grouped into three categories. It can be considered to be missing completely at random (does not depend on either the known values or the missing data), missing at random (may depend on the known values, but not on the value of the missing data itself) or missing not at random (dependent on the variable that is missing) (Batista & Monard, 2003). Importantly, each class of missingness requires specific MVI. Watson et al. (2017) gathered TL data for female adolescent soccer players and found no obvious differences in missing data with respect to date, age group, or specific individual participants. Anecdotally, missing TL data will most commonly occur because a player forgets to record it or experiences technical issues. Therefore, missing TL data is most likely missing completely at random and this must be taken into account when selecting a MVI method. This is supported by Horgan et al. (2020) who considered the missingness of TL data to be completely at random. The extent of missing data, defined as the percentage of cases with missing data for a given variable, will also have an impact on the robustness of the various MVI techniques (Fox-Wasylyshyn & El-Masri, 2005; Lin & Tsai, 2020). There is an abundance of literature that has examined the use of MVI in research (Batista & Monard, 2003; Collins, 2006; Fox-Wasylyshyn & El-Masri, 2005; Lin & Tsai, 2020); however, only one study to date has explored MVI methods in applied TL data. Benson et al. (2021) used single imputation and machine learning methods to impute the simulated missing TL data of youth basketball players and found that MVI methods specific to that session (i.e. team data), and methods that combine information about the session and the individual (i.e. machine learning models), both outperformed methods based solely on individual player data. However, their study did not examine the impact various levels of missingness have on the accuracy of the MVI methods, nor the impact missing data has on other TL metrics. Furthermore, research relating to TL may not be comparable between youth and adult populations and they need to be investigated separately (Watson et al., 2017).

There is currently no definitive best practice for addressing missing TL data in team sports and McCall et al. (2018) concluded that future research should explore methods of dealing with missing data. Therefore, the purpose of this study was to explore (1) various practical methods of MVI in team TL data across differing levels of missingness in order to determine the most appropriate method for practitioners in terms of use and accuracy and (2) the effect of missing data on various metrics (ACWR, training strain, training monotony) that are derived from sRPE data. Establishing a real-world practical approach to missing TL data may support practitioners in implementing TL monitoring strategies and in-turn, facilitate safe TL prescription that could potentially mitigate injury risk and enhance sporting performance.

Methods

Experimental approach

This study is a secondary analysis of data from a previously published study that investigated the association between TL Indices and injuries in elite soccer players (Tiernan et al., 2020). Thirty-two consecutive weeks of complete data collection took place over a soccer season. A member of the research team (CT) attended all training sessions and matches to ensure a complete dataset was achieved (i.e. no missing data was present after data collection). After familiarization and anchoring of the RPE scale were sRPE was recorded daily for each player resulting in 224 data points per player across the season.

Data were randomly removed at a range of 5–50% in increments of 5% and data were imputed using twelve MVI methods. The performance of each method was evaluated using appropriate error measures.

Subjects

Ten male professional soccer players volunteered to take part in the study (age, 24.8 ± 5.0 years; height, 181.2 ± 5.1 cm; mass, 78.7 ± 6.4 kg). All players were contracted and played for the senior first team, competing in the Premier League of Ireland. All subjects were informed of the benefits and risks of participating, and written informed consent was completed before undertaking the data collection in compliance with the Declaration of Helsinki. Ethical approval for this research was provided by the Institutional Research Ethics Review Board.

Procedures

Familiarization

Prior to data collection, subjects were given a familiarization period of one week which consisted of 3 separate education sessions on the 0–10 Rating of Perceived Exertion (RPE) scale (Foster et al., 2001). Additionally, subjects completed a yo-yo intermittent test in order to anchor the top end of the RPE scale and help establish accurate RPE scores for exercise intensity (Gearhart, 2008). As this is a maximal test, subjects were instructed afterwards that their last run should equate to the highest score in the RPE scale.

Training load data collection

A typical training week consisted of four team training days, a competitive match and two rest/recovery days. The composition of training days varied but consisted of fitness conditioning, skill specific training, tactics and match preparation, resistance training, speed and agility training and conditioned games. Internal TL was calculated using the sRPE method as described by Foster et al. (2001) (RPE x session duration in minutes). sRPE has been shown to be a very common, practical, reliable and valid method of measuring internal TL across a wide variety of exercise modalities including soccer (Chen et al., 2002; Foster et al., 2001; Impellizzeri et al., 2004; Lovell et al., 2013). The subjects recorded the duration of the session they completed (in minutes) and their RPE measured on a chromatic coloured 0–10 scale (Serafim et al., 2014). sRPE was recorded using a specifically designed mobile phone app created using Google Forms (Google, Mountain View, California, USA) and AppSheet (AppSheet Seattle, Washington, USA). The app was installed on the players' phones, to avoid external influences (Scantlebury et al., 2018) (Figure 1). If multiple sessions were completed in a day, the sRPE loads were summed at the end of the day to give a daily TL (Williams et al., 2017a). Recovery days were noted as 0 AU (Esmaeili et al., 2018).

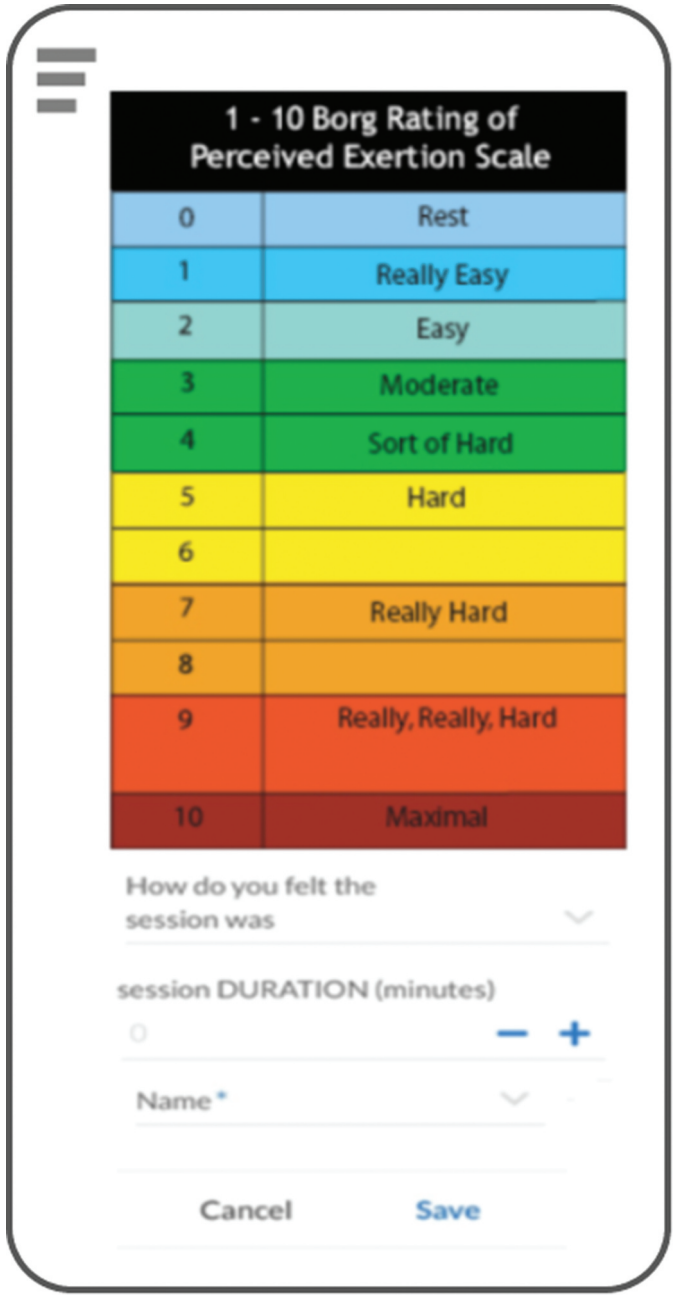

Figure 1. Screenshot of the specifically designed mobile phone app used by the subjects to record their internal training load.

Data removal and missing value imputation

Data were randomly removed from the complete TL dataset (2240 data points) in increments of 5% ranging from 5% to 50% to represent missing data. This was completed using the Microsoft Excel random function (RAND) whereby each TL data point was assigned a random number between 1–100 and then the required percentage of missingness was highlighted and removed. Data imputation involved replacing the randomly removed missing TL values with estimated ones using the various imputation methods. Many of the MVI methods (e.g., regression imputation) supported in the literature were not applicable to sRPE data as their calculation would require other variables that are related to sRPE. Additionally, it was important that MVI methods aligned with the study aims of exploring methods that are practical in a real-world context. Many MVI methods require specialist software that are not freely available to practitioners working with teams.

Therefore, the twelve MVI methods were theorized and examined based on their structure and their practicality (i.e. can be feasibly used by a practitioner working with a soccer team). A description of all the MVI methods used is presented in Table 1. For all imputation methods, previously imputed data were not included in any further calculations within that method.

Additional training load metrics

Additional metrics derived from sRPE data were calculated, and illustrated, to demonstrate the impact missing data has on their computation. These additional metrics are commonly used in applied practice. Training monotony is a measure of day-to-day training variability during a training week and was calculated by dividing the mean session TL by the standard deviation of the TL over a 1-week period (Foster, 1998). Training strain is a value that represents the overall stress that the athlete was exposed to throughout the training week and was calculated by multiplying the total weekly TL by the training monotony score (Foster, 1998). The acute:chronic workload ratio (ACWR) is a relative measure of TL which takes into account the current workload (acute) and the workload that an athlete has been prepared for (chronic) (Griffin et al., 2020a). It was calculated using the rolling average model (RA) and the exponentially weighted moving average model (EWMA). The coupled RA ACWR was calculated by dividing the acute TL by the chronic TL. The EWMA ACWR was calculated as described by Williams et al. (2017b). An acute period of 7-days and a chronic period of 28-days were used for both ACWR models.

Statistical analysis

Descriptive statistics were computed for the complete TL data as median and interquartile range (IQR). The Normalized Root Mean Squared Error (NRMSE) and the Mean of Absolute Deviations (MAD), standard statistical metrics to measure model performance, were used to compare the performance of the MVI methods across the levels of missingness. A zero value of NSMSE and MAD would indicate a perfect fit to the data,; therefore,lower values on these metrics indicate better accuracy.

Normalized root-mean-square error (NRMSE):

$$NRMSE = \frac{\sqrt{\frac{1}{m}(y_i - \widehat{y_i})^2}}{IQR}$$

where m = number of imputed values, y_i = true value, $\widehat{y_i}$ = imputed value and *IQR* = Interquartile range of y_i.

The Mean of Absolute Deviations (MAD) is the mean of the absolute deviations between the complete dataset and the dataset containing the imputed values as a measure of the average error introduced by the imputed data.

Mean of Absolute Deviations (MAD):

$$MAD = \frac{\sum |y_i - \hat{y}_i|}{n}$$

where n = sample size.

The effects of the imputed data on the ACWR, monotony and strain metrics derived from TL are illustrated graphically.

Table 1. Explanation of the missing value imputation methods.

Name	Explanation
Team data derived	
Daily Team Mean (DTMean)	The mean of all the other players' TL for that particular session.
Daily Team Median (DTMedian)	The median of all the other players' TL for that particular session.
Player data derived	
Player Mean (PM)	The mean TL of all the individual player's previous training sessions.
Player Median	The median TL of all the individual player's previous training sessions.
R4W Player Mean	The rolling mean TL of all the individual player's previous 4 weeks. This method addresses the difference in training practices depending on the time of season (e.g., much larger volume in pre-season compared to in-season).
R4W Player Median	The rolling median TL of all the individual player's previous 4 weeks.
Day of Week Mean	The mean TL of all the individual player's previous training sessions that fall on the same day of the week (i.e. if data on a Monday are missing then they are replaced with the mean TL of all the previous Mondays that were not missing).
Day of Week Median	The median TL of all the individual player's previous training sessions that fall on the same day of the week.
Mixed team and player data derived	
Mixed Mean	Accounts for both the player's perception of the previous TL and their teammates' perception of that particular session. It gives both mean values an equal weighing. **((player mean of all previous sessions + daily team mean)/2)**
Mixed Median	Accounts for both the player's perception of the previous TL and their teammates' perception of that particular session. It gives both median values an equal weighing.
Mixed R4W Mean	Accounts for both the player's perception of the recent TL (past 4 weeks) and their teammates' perception of that particular session. Justification for using the 4-week period is that it aligns well with the periodization strategies used in many team sports (Williams et al., 2017b). It gives both mean values an equal weighing. **((player mean of the previous 4 weeks + daily team mean)/2)**
Mixed R4W Median	Accounts for both the player's perception of the recent TL (past 4 weeks) and their teammates' perception of that particular session. It gives both median values an equal weighing. **((player median of the previous 4 weeks + daily team median)/2)**

Performance analysis of the imputation methods was undertaken in the R software environment, version 4.0.3 (R Core Team, 2020; Dowle & Srinivasan, 2020; Wickham et al., 2019; Erdely & Castillo, 2017; Zeileis and Grothendieck, 2005).

Results

Descriptive analysis

A total of 2240 player day exposures were included in the analysis of this study, resulting in a median daily TL of 180 AU (IQR = 450 AU). Days off (TL = 0 AU) accounted for 36.3% or 813 days. Training days (n = 1151, 51.4%) resulted in a median TL of 300 AU (IQR = 345 AU) while match days (n = 276, 12.3%) resulted in a median TL of 568 AU (IQR = 460).

Accuracy results

The best-fitting MVI method was Daily Team Mean (DTMean) with NRMSE ranging from 0.42 to 0.46 across all levels of missingness (Table 2 and Figure 2(a)). The MVI methods of Daily Team Median (DTMedian), Mixed Mean, Mixed Median, Mixed R4W Mean and Mixed R4W Median produced similar NRMSE values of 0.44–0.49. The methods of Player Mean (PM), Player Median, R4W Player Mean, R4W Player Median, Day of Week Mean and Day of Week Median were each a poorer fit with NRMSE values of 0.48–0.68.

For all imputation methods, as the percentage of missing data increases so too does the MAD in a linear manner (Table 3 and Figure 2(b)). At 5% missingness, the mean MAD value of all MVI methods is 9.2 while at 50% missingness this increased to 90.8. Figure 3 illustrates the impact that the level of data missingness has on the fit of the MVI method by comparing 10 and 20% missingness after DTMean MVI imputation.

At 10% missingness, the DTMean MVI method resulted in a NRMSE of 0.43 whereas the PM MVI method produced a NRMSE of 0.56. As an illustration of the difference in accuracy between these two methods, Figure 4 presents a comparison of the imputed values from both these MVI methods and the corresponding complete data values for the missing days from a randomly selected single subject.

Table 2. Normalized root-mean-square error (NRMSE) values for the different methods of missing value imputation (5–50% missingness).

	Percentage Missingness (%)									
MVI Method	5	10	15	20	25	30	35	40	45	50
Daily Team Mean	0.46	0.43	0.42	0.46	0.43	0.45	0.46	0.45	0.43	0.46
Daily Team Median	0.49	0.45	0.44	0.48	0.46	0.47	0.49	0.47	0.44	0.49
Player Mean	0.60	0.56	0.57	0.59	0.60	0.57	0.58	0.63	0.58	0.60
Player Median	0.65	0.59	0.60	0.63	0.63	0.61	0.60	0.67	0.62	0.63
R4W Player Mean	0.59	0.56	0.58	0.59	0.60	0.57	0.58	0.62	0.58	0.60
R4W Player Median	0.65	0.60	0.62	0.64	0.65	0.61	0.61	0.68	0.63	0.64
Day of Week Mean	0.54	0.51	0.48	0.52	0.54	0.53	0.50	0.55	0.50	0.52
Day of Week Median	0.58	0.54	0.53	0.56	0.56	0.56	0.53	0.59	0.53	0.55
Mixed Mean	0.48	0.44	0.45	0.47	0.47	0.46	0.46	0.49	0.46	0.47
Mixed Median	0.48	0.44	0.44	0.47	0.46	0.45	0.46	0.49	0.45	0.48
Mixed R4W Mean	0.48	0.44	0.45	0.47	0.47	0.46	0.46	0.48	0.45	0.47
Mixed R4W Median	0.48	0.44	0.45	0.47	0.47	0.46	0.46	0.49	0.45	0.48

Projected effects on training load metrics

Figure 5 illustrates the effect missing data had on daily training metrics derived from the sRPE TL data for a randomly selected player, and the improvement in the accuracy of the metrics by using the DTMean MVI. The computed metrics are the ACWR using the exponentially weighted moving average model (A) and ACWR using the coupled rolling average model (B), training monotony (C) and training strain (D).

Discussion

This study aimed to evaluate practical methods of addressing missing data in TL monitoring datasets in order to determine the most suitable MVI method to be used by strength and conditioning coaches and sport scientists working in a team sport environment. The median daily TL of the 1151 training days was 300 AU (IQR = 345 AU). This is similar to the results of previous research that found daily TL of professional soccer players was 379.9 ± 198.3 AU (Moalla et al., 2016). The daily TL fluctuates to the extent that when rest days are excluded, a bimodal distribution is seen. This is typical as in-season TL prescription of professional soccer players is influenced by competition frequency resulting in a progressive decrease in TL prior to match day (Kelly et al., 2020). Additionally, TL tends to be greatest on match day compared to all other days of the week (Kelly et al., 2020). This is supported by the data collected

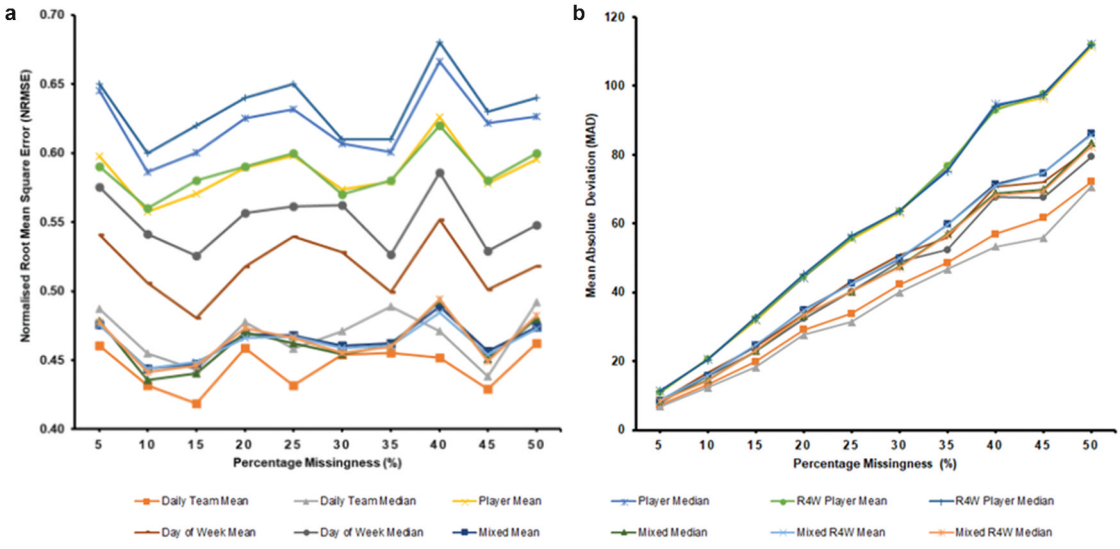

Figure 2. Comparison of the various missing value imputation methods in terms of NRMSE (a) and MAD (b) across the missing value percentages. Note: An explanation of each data imputation methods is provided in.Table 1.

Table 3. Mean absolute deviation (MAD) values for the different methods of missing value imputation (5–50% missingness).

	Percentage Missingness (%)									
MVI Method	5	10	15	20	25	30	35	40	45	50
Daily Team Mean	07.3	13.2	19.9	29.1	33.8	42.3	48.6	56.9	61.6	72.1
Daily Team Median	06.8	12.3	18.2	27.6	31.4	40.0	46.7	53.3	55.9	70.6
Player Mean	10.9	20.7	31.8	44.3	55.4	63.1	76.7	93.4	96.4	111.3
Player Median	11.4	20.6	32.2	44.3	56.0	63.7	75.7	94.7	97.1	112.1
R4W Player Mean	10.9	20.7	32.3	44.6	55.9	63.6	76.8	92.9	97.6	111.9
R4W Player Median	11.3	20.4	32.8	45.2	56.5	63.6	75.2	94.0	97.5	112.0
Day of Week Mean	08.3	16.6	24.3	33.7	43.2	50.9	56.0	70.7	71.9	82.3
Day of Week Median	07.9	15.7	22.9	32.3	40.3	48.9	52.4	67.7	67.5	79.3
Mixed Mean	08.6	15.9	24.6	34.9	42.5	49.8	59.8	71.4	74.6	86.1
Mixed Median	08.4	14.6	22.9	33.3	40.3	47.6	57.0	68.9	70.0	83.4
Mixed R4W Mean	08.6	15.8	24.8	34.8	42.6	50.0	59.6	70.9	74.6	85.7
Mixed R4W Median	08.3	14.4	23.1	33.5	40.3	47.3	56.6	68.2	69.4	82.5

in the current study as the median TL of the 276 match days was 568 AU (IQR = 460 AU).

The main finding of the current study is that the DTMean had the highest accuracy of the MVI methods examined with NRMSE values across the missing percentages from 5–50% ranging from 0.42 to 0.46. As evident from Table 2 and Figure 2(a), DTMean MVI resulted in data that was most closely related to the observed data, regardless of the extent of missingness. It is also evident that the imputation methods DTMedian, Mixed Mean, Mixed Median, Mixed R4W Mean and Mixed R4W Median achieved similar results in terms of NRMSE (0.44–0.49) and performed only slightly poorer than DTMean. It appears that these six MVI methods may all be effective at addressing missing TL data. In fact, in terms of MAD, the DTMedian performed best with values ranging from 7.3–72.1 across the missing percentages while DTMean resulted in MAD values ranging from 6.8–70.6 (Table 3 and Figure 2(b)). This would indicate that MVI should incorporate the daily team value (mean or median). However, based on scores of both the NRMSE and the MAD, the DTMean is the most accurate MVI method. The remaining six MVI methods (Day of Week Mean, Day of Week Median, Player Mean, Player Median, R4W Player Mean and R4W Player Median) present an increase in NRMSE values (0.48–0.67) suggesting they are inferior methods.

Missing TL data are inherent within applied research conducted on elite sports teams (Jaspers et al., 2018; Windt et al., 2018). In instances of missing sRPE data, McCall et al. (2018) imputed the player's seasonal mean TL. The decision to use this method was derived by means of a round table discussion with authors and practitioners. Similarly, Sampson et al. (2018) calculated the player's weekly TL average and added it to the dataset. Both McCall et al. (2018) and Sampson et al. (2018) address missing data MVI methods based on a player's pervious recorded data, similar to the PM methods examined in this study. Performance evaluation of the MVI methods in this study found MVI methods based on daily team data outperformed methods based on a player's data, with the DTMean

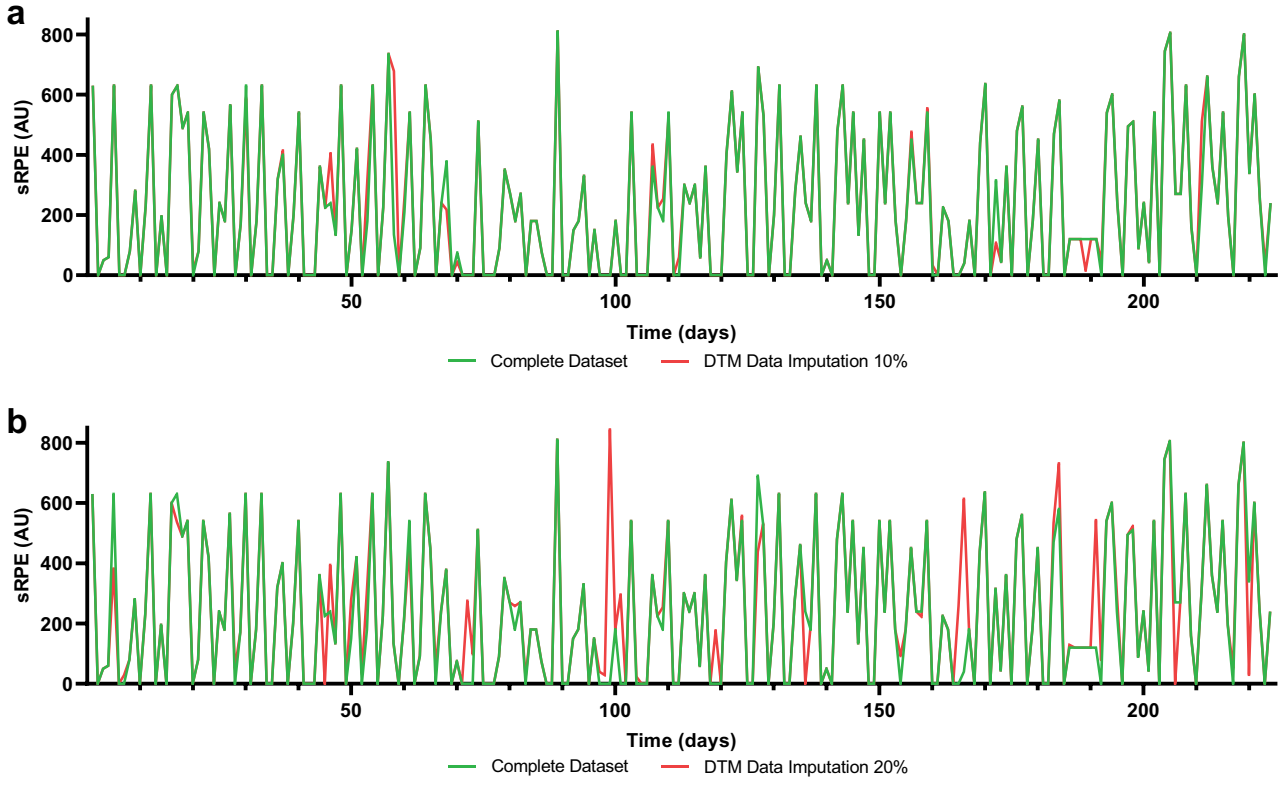

Figure 3. The complete sRPE dataset and the dataset post Daily Team Mean (DTMean) missing value imputation when missingness is 10% (a) and 20% (b). A randomly selected single player's data across a full season is represented.

Figure 4. Player Mean (PM) and Daily Team Mean (DTMean) imputed sRPE values at 10% missingness compared to the corresponding complete values for a randomly selected player across a full season.

MVI method having the lowest NRMSE errors across all levels of missingness. This is similar to the findings of Benson et al. (2021). As illustrated in Figure 4, the DTMean is a better fit to the observed data compared to the PM MVI method, across nearly all of the missing days of the subject's sRPE data. It would appear from this that the PM method fails to account for the fluctuations in day-to-day TL. It is also evident from Figure 4 that, on a small number of days, the DTMean MVI method differed from the true value. For instance, on day 6 the DTMean MVI method overestimated the true value by 543 AU. The reason for this inaccuracy is likely due to it being a - match day and the particular subject only played for a short duration while the other 9 players in the dataset played a full 90 minutes. This highlights the need for coaches to exercise caution and use all available context around the missed session when using this method of MVI.

The distribution of TL frequency explains the oscillating nature of TL and why MVI imputation methods using the player's mean TL may be inaccurate. This is supported by the NRMSE values for both methods (Table 1 and Figure 2(a)). McCall et al. (2018) suggest that player mean TL reflects the real-world setting as it is commonly used in practice by teams. The results of the current study offer an alternative method that may provide practitioners with a more accurate method of addressing missing data in the DTMean. Furthermore, the DTMean method does not require data collection from any previous sessions whereas a number of the other methods do. This highlights its practicality and potential to be integrated into the real-world practices of practitioners working with teams.

Unsurprisingly, as the percentage of missing data increases, the accuracy of each MVI method decreases (Table 3). There appears to be a linear relationship between the amount of missing data present and the accuracy of the MVI methods. This is illustrated in Figure 2(b) as MAD values increase with the percentage of missingness. There is limited research regarding the level of missingness that is common in team sport. Watson et al. (2017) had a compliance rate of 84.7% overall when gathering TL data with a female adolescent soccer players across a full season, however, it should be noted that wellbeing data was also gathered. This may have increased the occurrence of missing data as measures of wellbeing have been found to achieve a lower compliance rate of 70% in professional Australian footballers. McLaren et al. (2018) collected sRPE data across an eight-week pre-season training period in a squad of thirty-five professional rugby union players and achieved 100% compliance. Outside of team sports, sRPE compliance rates have been found to be 100% with endurance athletes (Roos et al., 2018). Therefore, it can be assumed that it is unlikely missing sRPE data will exceed 10% (missing one in ten sessions).

When the DTMean was imputed at 10% missingness, the NRMSE value was 0.43. The typical training week of the subjects in the current study consisted of 4 training days, a competitive match and 2 rest days, resulting in a mean weekly TL of 1696 AU. Missing sRPE data for one in ten training/competition days (10%) equates to one session every 2 weeks (3392 AU). If a practitioner replaces missing data using the DTMean MVI method they can be reasonably confident that it will be close to the true value and, in turn, will likely aid TL prescription. Figure 3 demonstrates the practical impact the level of missingness has on the accuracy of the DTMean MVI method. At 10% missingness, the imputed values appear much more closely related to the observed data compared to when missingness is 20%. This is supported by the MAD values of 13.2 at 10% missingness and 29.1 at 20%. Practitioners can be relatively confident that when missingness is restricted to 10% or less, then DTMean MVI will accurately replace the missing values. If missingness does exceed 10%, it is suggested practitioners re-examine their TL monitoring techniques. The level of missingness that renders the effectiveness of MVI questionable may be sport, or even team, dependent. However, due to the linear

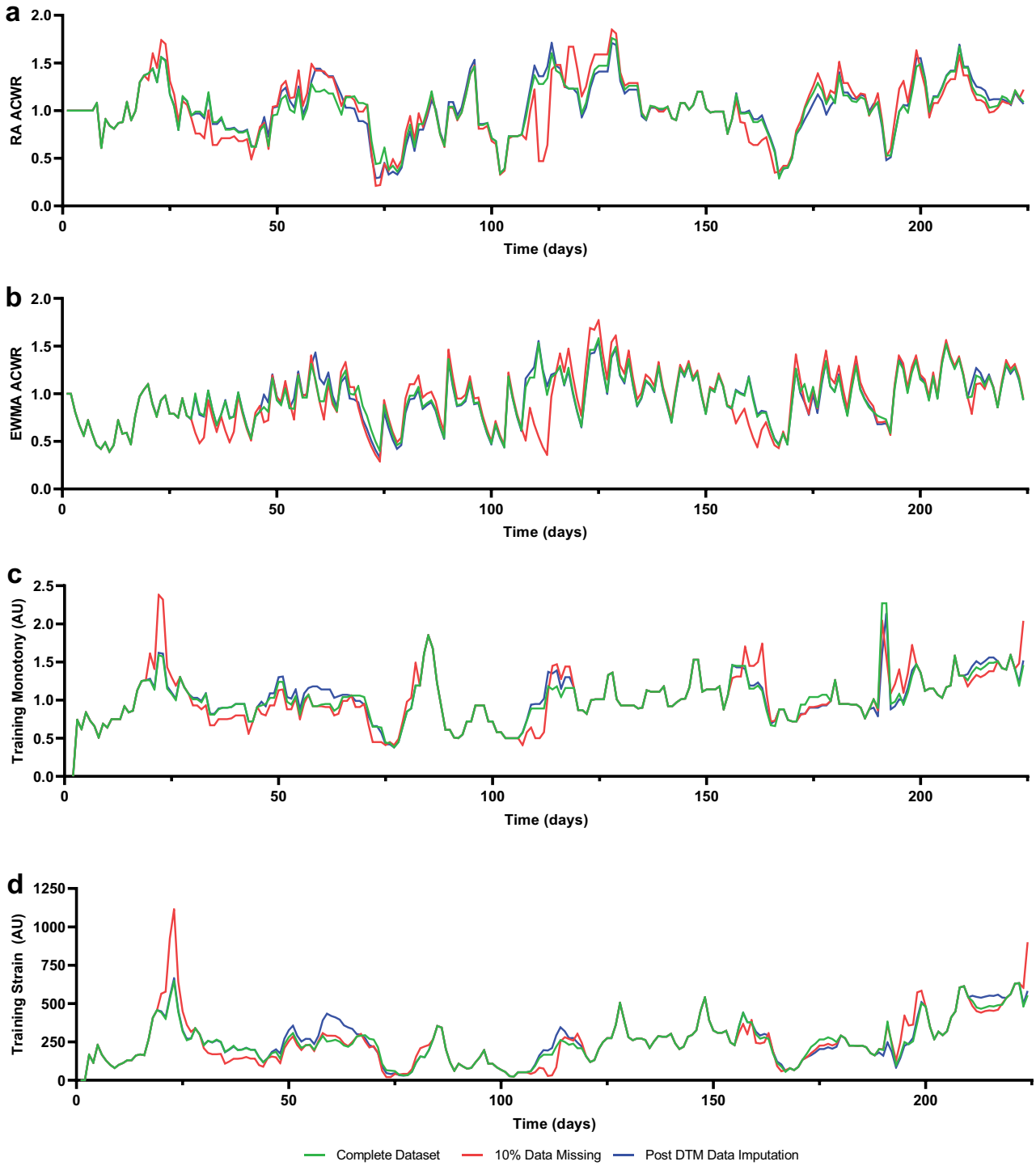

Figure 5. The effect of missing data and Daily Team Mean (DTMean) missing value imputation on various metrics of training load: Rolling average acute:chronic workload ratio (a), exponentially weighted moving average acute:chronic workload ratio (b), training monotony (c) and training strain (d). A randomly selected single player's data across a full season are represented.

relationship between missingness and accuracy, missing data should always be kept to a minimum. To ensure compliance rate is as high as possible, practitioners should provide players with regular feedback and evidence that the data are informing their training (Griffin et al., 2020c).

It is common practice in research to exclude subjects with incomplete datasets from analysis or to use methods of statistical analyses that are robust to missing and unbalanced data (Windt et al., 2018). This method, however, is not applicable to practitioners working with teams in the real-world as they require systematic TL data for all players to aid future safe TL prescription. Not addressing missing sRPE data may lead to inaccurate calculations of other TL metrics (e.g., ACWR, training monotony, training strain). Figure 5 demonstrates the negative effect of ignoring missing sRPE data prior to calculating further metrics derived from the TL. This may result in a causal

sequence whereby practitioners prescribe inappropriate TL and consequently increase injury risk and hinder sports performance (Soligard et al., 2016). It is evident from Figure 5 that these metrics will likely be more accurate when DTMean MVI is used prior to their calculation. It is therefore recommended that missing TL data be addressed using the DTMean prior to the calculation of these TL metrics.

A limitation of the DTMean MVI method is that it is not applicable outside of a team environment. For instance, if a player is in post-injury rehabilitation before returning to team activities then the team mean TL should not be used. Future research is required to examine the best method of MVI in this scenario. Additionally, there is a need to examine MVI in the context of individual sports where it is not possible to calculate team means. Research addressing the issue of missing TL data is still in its infancy and further examinations into other sports, levels of play, larger datasets and alternative imputation methods are still needed. In soccer, there is little meaningful difference between the TL across playing positions (Kelly et al., 2019) but this may not be the case across all team sports. The DTMean MVI method appears to be practical but this does not confirm its translation into the real-world environment. To fully determine its practicality and ability to be integrated into practice, future research would be required that investigates this. There is a need to examine practitioners' ratings of the MVI methods and their practicality in relation to sample size, recording, spreadsheet, and analysis procedures. Additionally, future research could aim to develop software packages that utilize the more sophisticated methods of MVI (e.g., hot and cold decking, expectation maximization, multiple imputation). However, what is critical here is that practicality and end-user friendliness are prioritized. This study, however, is the first to examine the impact of missing TL on various metrics (ACWR, training monotony and strain) and offers a practical and effective method of MVI to address it.

Practical applications

Practitioners should strive to keep missing data at a minimum. However, when collecting TL data longitudinally, missing data is inevitable due to many factors. The presence of missing sRPE data may lead to inaccurate calculations of further TL metrics and in turn, poor TL prescription. The DTMean MVI method offers practitioners, working with teams, a practical and effective method of addressing the negative consequences of missing TL data. The DTMean MVI method appears to be easy to use, quick to calculate, does not require specialist software or previous data collection. As the monitoring of TL is becoming more and more prevalent, the findings of this study address an ever increasing challenge for practitioners working with team sport athletes.

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Data availability

The analyzed datasets and the R code used during the current study are available from the corresponding author on reasonable request.

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All authors contributed to the design and implementation of the research, to the analysis of the results and to the writing of the manuscript. All authors have read and agreed to the published version of the manuscript.

References

Akenhead, R., & Nassis, G. P. (2016). Training load and player monitoring in high-level football: Current practice and perceptions. *International Journal of Sports Physiology and Performance*, *11*(5), 587–593. https://doi.org/10.1123/ijspp.2015-0331

Bahr, R., & Krosshaug, T. (2005). Understanding injury mechanisms: A key component of preventing injuries in sport. *British Journal of Sports Medicine*, *39*(6), 324–329. https://doi.org/10.1136/bjsm.2005.018341

Ball, S., Halaki, M., Sharp, T., & Orr, R. (2018). Injury patterns, physiological profile, and performance in university rugby union. *International Journal of Sports Physiology and Performance*, *13*(1), 69–74. https://doi.org/10.1123/ijspp.2017-0023

Batista, G. E., & Monard, M. C. (2003). An analysis of four missing data treatment methods for supervised learning. *Applied Artificial Intelligence*, *17*(5–6), 519–533. https://doi.org/10.1080/713827181

Benson, L. C., Stilling, C., Owoeye, O. B., & Emery, C. A. (2021). Evaluating methods for imputing missing data from longitudinal monitoring of athlete workload. *Journal of Sports Science & Medicine*, *20*, 187–195. https://doi.org/10.52082/jssm.2021.188

Borg, G. A. (1982). Psychophysical bases of perceived exertion. *Medicine & Science in Sports & Exercise*, *14*(5), 377–381. https://doi.org/10.1249/00005768-198205000-00012

Chen, M. J., Fan, X., & Moe, S. T. (2002). Criterion-related validity of the Borg ratings of perceived exertion scale in healthy individuals: A meta-analysis. *Journal of Sports Sciences*, *20*(11), 873–899. https://doi.org/10.1080/026404102320761787

Collins, L. M. (2006). Analysis of longitudinal data: The integration of theoretical model, temporal design, and statistical model. *Annual Review of Psychology*, *57*(1), 505–528. https://doi.org/10.1146/annurev.psych.57.102904.190146

Comyns, T., & Hannon, A. (2018). Strength and conditioning coaches' application of the session rating of perceived exertion method of monitoring within professional rugby union. *Journal of Human Kinetics*, *61*(1), 155–166. https://doi.org/10.1515/hukin-2017-0118

Dowle, M., & Srinivasan, A. (2020). *Data.table: Extension of data.frame* (R package version 1.13.6). Retrieved from https://cran.r-project.org/package=data.table

Drew, M. K., & Finch, C. F. (2016). The relationship between training load and

injury, illness and soreness: A systematic and literature review. *Sports Medicine*, *46*(6), 861–883. https://doi.org/10.1007/s40279-015-0459-8

Drew, M. K., Raysmith, B. P., & Charlton, P. C. (2017). Injuries impair the chance of successful performance by sportspeople: A systematic review. *British Journal of Sports Medicine*, *51*(16), 1209–1214. https://doi.org/10.1136/bjsports-2016-096731

Eckard, T. G., Padua, D. A., Hearn, D. W., Pexa, B. S., & Frank, B. S. (2018). The relationship between training load and injury in athletes: A systematic review. *Sports Medicine*, *48*(8), 1929–1961. https://doi.org/10.1007/s40279-018-0951-z

Erdely, A., & Castillo, I. (2017). *Cumstats: Cumulative descriptive statistics* (R Package version 1.0). Retrieved from https://CRAN.R-project.org/package=cumstats

Esmaeili, A., Hopkins, W. G., Stewart, A. M., Elias, G. P., Lazarus, B. H., & Aughey, R. J. (2018). The individual and combined effects of multiple factors on the risk of soft tissue non-contact injuries in elite team sport athletes. *Frontiers in Physiology*, *9*, 1280. https://doi.org/10.3389/fphys.2018.01280

Foster, C. (1998). Monitoring training in athletes with reference to overtraining syndrome. *Medicine & Science in Sports & Exercise*, *30*(7), 1164–1168. https://doi.org/10.1097/00005768-199807000-00023

Foster, C., Florhaug, J. A., Franklin, J., Gottschall, L., Hrovatin, L. A., Parker, S., Doleshal, P., & Dodge, C. (2001). A new approach to monitoring exercise training. *Journal of Strength and Conditioning Research*, *15*(1), 109–115. https://doi.org/10.1519/00124278-200102000-00019

Fox-Wasylyshyn, S. M., & El-Masri, M. M. (2005). Handling missing data in self-report measures. *Research in Nursing & Health*, *28*(6), 488–495. https://doi.org/10.1002/nur.20100

Gearhart, R. F., Jr. (2008). Ratings of perceived exertion and oxygen consumption during maximal, graded, treadmill exercise following different anchoring procedures. *European Journal of Sport Science*, *8*(1), 35–40. https://doi.org/10.1080/17461390701819352

Gouttebarge, V., Tol, J. L., & Kerkhoffs, G. M. (2016). Epidemiology of symptoms of common mental disorders among elite Gaelic athletes: A prospective cohort study. *The Physician and Sportsmedicine*, *44*(3), 283–289. https://doi.org/10.1080/00913847.2016.1185385

Griffin, A., Kenny, I. C., Comyns, T. M., & Lyons, M. (2020a). The association between the acute: Chronic workload ratio and injury and its application in team sports: A systematic review. *Sports Medicine*, *50*(3), 561–580. https://doi.org/10.1007/s40279-019-01218-2

Griffin, A., Kenny, I. C., Comyns, T. M., & Lyons, M. (2020b). Training load monitoring in amateur Rugby Union: A survey of current practices. *Journal of Strength and Conditioning Research*. Epub ahead of print. https://doi.org/10.1519/JSC.0000000000003637 .

Griffin, A., Kenny, I. C., Comyns, T. M., & Lyons, M. (2020c). The development and evaluation of a training monitoring system for amateur Rugby Union. *Applied Sciences*, *10*(21), 7816. https://doi.org/10.3390/app10217816

Herman, L., Foster, C., Maher, M. A., Mikat, R. P., & Porcari, J. P. (2006). Validity and reliability of the session RPE method for monitoring exercise training intensity. *South African Journal of Sports Medicine*, *18*(1), 14–17. https://doi.org/10.17159/2078-516X/2006/v18i1a247

Horgan, B. G., Drew, M. K., Halson, S. L., Piromalli, L. E., Drinkwater, E. J., Chapman, D. W., & Haff, G. G. (2020). Impaired recovery is associated with increased injury and illness: A retrospective study of 536 female netball athletes. *Scandinavian Journal of Medicine & Science in Sports*, *31* (3), 691–701. Epub ahead of print. https://doi.org/10.1111/sms.13866

Impellizzeri, F. M., Rampinini, E., Coutts, A. J., Sassi, A., & Marcora, S. M. (2004). Use of RPE-based training load in soccer. *Medicine & Science in Sports & Exercise*, *36*(6), 1042–1047. https://doi.org/10.1249/01.MSS.0000128199.23901.2F

Jaspers, A., Kuyvenhoven, J. P., Staes, F., Frencken, W. G. P., Helsen, W. F., & Brink, M. S. (2018). Examination of the external and internal load indicators' association with overuse injuries in professional soccer players. *Journal of Science and Medicine in Sport*, *21*(6), 579–585. https://doi.org/10.1016/j.jsams.2017.10.005

Kelly, D. M., Strudwick, A. J., Atkinson, G., Drust, B., & Gregson, W. (2020). Quantification of training and match-load distribution across a season in elite English Premier League soccer players. *Science and Medicine in Football*, *4*(1), 59–67. https://doi.org/10.1080/24733938.2019.1651934

Lin, W. C., & Tsai, C. F. (2020). Missing value imputation: A review and analysis of the literature (2006–2017). *Artificial Intelligence Review*, *53* (2), 1487–1509. https://doi.org/10.1007/s10462-019-09709-4

Lovell, T. W., Sirotic, A. C., Impellizzeri, F. M., & Coutts, A. J. (2013). Factors affecting perception of effort (session rating of perceived exertion) during Rugby league training. *International Journal of Sports Physiology and Performance*, *8*(1), 62–69. https://doi.org/10.1123/ijspp.8.1.62

Manley, G., Gardner, A. J., Schneider, K. J., Guskiewicz, K. M., Bailes, J., Cantu, R. C., Castellani, R. J., Turner, M., Jordan, B. D., Randolph, C., Dvořák, J., Hayden, K. A., Tator, C. H., McCrory, P., & Iverson, G. L. (2017). A systematic review of potential long-term effects of sport-related concussion. *British Journal of Sports Medicine*, *51*(12), 969–977. https://doi.org/10.1136/bjsports-2017-097791

McCall, A., Dupont, G., & Ekstrand, J. (2018). Internal workload and non-contact injury: A one-season study of five teams from the UEFA Elite Club Injury Study. *British Journal of Sports Medicine*, *52*(23), 1517–1522. https://doi.org/10.1136/bjsports-2017-098473

McCrory, P., Meeuwisse, W., Dvořák, J., Aubry, M., Bailes, J., Broglio, S., Cantu, R. C., Cassidy, D., Echemendia, R. J., Castellani, R. J., Davis, G. A. (2017). Consensus statement on concussion in sport—the 5th international conference on concussion in sport held in Berlin. *British Journal of Sports medicine*, *51*(11), 838–847. https://doi.org/10.1136/bjsports-2017-097699

McLaren, S. J., Smith, A., Bartlett, J. D., Spears, I. R., & Weston, M. (2018). Differential training loads and individual fitness responses to pre-season in professional rugby union players. *Journal of Sports Sciences*, *36*(21), 2438–2446. https://doi.org/10.1080/02640414.2018.1461449

Meeuwisse, W. H. (1994). Assessing causation in sport injury: A multifactorial model. *Clinical Journal of Sport Medicine*, *4*(3), 166–170. https://doi.org/10.1097/00042752-199407000-00004

Moalla, W., Fessi, M. S., Farhat, F., Nouira, S., Wong, D. P., & Dupont, G. (2016). Relationship between daily training load and psychometric status of professional soccer players. *Research in Sports Medicine*, *24*(4), 387–394. https://doi.org/10.1080/15438627.2016.1239579

Newgard, C. D., & Lewis, R. J. (2015). Missing data: How to best account for what is not known. *JAMA*, *314*(9), 940–941. https://doi.org/10.1001/jama.2015.10516

R Core Team. (2020). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing.

Rice, S. M., Purcell, R., De Silva, S., Mawren, D., McGorry, P. D., & Parker, A. G. (2016). The mental health of elite athletes: A narrative systematic review. *Sports Medicine*, *46*(9), 1333–1353. https://doi.org/10.1007/s40279-016-0492-2

Roos, L., Taube, W., Tuch, C., Frei, K., & Wyss, T. (2018). Factors that influence the rating of perceived exertion after endurance training. *International Journal of Sports Physiology and Performance*, *13*(8), 1042–1049. https://doi.org/10.1123/ijspp.2017-0707

Sampson, J. A., Murray, A., Williams, S., Halseth, T., Hanisch, J., Golden, G., & Fullagar, H. H. K. (2018). Injury risk-workload associations in NCAA American college football. *Journal of Science and Medicine in Sport*, *21* (12), 1215–1220. https://doi.org/10.1016/j.jsams.2018.05.019

Saw, A. E., Main, L. C., & Gastin, P. B. (2016). Monitoring the athlete training response: Subjective self-reported measures trump commonly used objective measures: A systematic review. *British Journal of Sports Medicine*, *50*(5), 281–291. https://doi.org/10.1136/bjsports-2015-094758

Scantlebury, S., Till, K., Sawczuk, T., Phibbs, P., & Jones, B. (2018). Validity of retrospective session rating of perceived exertion to quantify training load in youth athletes. *Journal of Strength and Conditioning Research*, *32* (7), 1975–1980. https://doi.org/10.1519/JSC.0000000000002099

Serafim, T. H., Tognato, A. C., Nakamura, P. M., Queiroga, M. R., Pereira, G., Nakamura, F. Y., & Kokubun, E. (2014). Development of the color scale of perceived exertion: Preliminary validation. *Perceptual and Motor Skills*, *119*(3), 884–900. https://doi.org/10.2466/27.06.PMS.119c28z5

Soligard, T., Schwellnus, M., Alonso, J.-M., Bahr, R., Clarsen, B., Dijkstra, H. P., Gabbett, T., Gleeson, M., Hägglund, M., Hutchinson, M. R., Janse Van Rensburg, C., Khan, K. M., Meeusen, R., Orchard, J. W., Pluim, B. M., Raftery, M., Budgett, R., & Engebretsen, L. (2016). How much is too much? (Part 1) International Olympic Committee consensus statement on load in sport and risk of injury. *British Journal of Sports Medicine*, *50* (17), 1030–1041. https://doi.org/10.1136/bjsports-2016-096581

Taylor, K., Chapman, D., Cronin, J., Newton, M. J., & Gill, N. (2012). Fatigue monitoring in high performance sport: A survey of current trends. *The Journal of Australian Strength and Conditioning*, *20*, 12–23.

Tiernan, C., Comyns, T., Lyons, M., Nevill, A. M., & Warrington, G. (2020). The association between training load indices and injuries in elite soccer players. *Journal of Strength and Conditioning Research*. Epub ahead of print. https://doi.org/10.1519/jsc.0000000000003914 .

Watson, A., Brickson, S., Brooks, A., & Dunn, W. (2017). Subjective well-being and training load predict in-season injury and illness risk in female youth soccer players. *British Journal of Sports Medicine*, *51*(3), 194–199. https://doi.org/10.1136/bjsports-2016-096584

Weston, M., Siegler, J., Bahnert, A., McBrien, J., & Lovell, R. (2015). The application of differential ratings of perceived exertion to Australian Football League matches. *Journal of Science and Medicine in Sport*, *18*(6), 704–708. https://doi.org/10.1016/j.jsams.2014.09.001

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T., Miller, E., Bache, S., Müller, K., Ooms, J., Robinson, D., Seidel, D., Spinu, V., Takahashi, K., & Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, *4*(43), 1686. https://doi.org/10.21105/joss.01686

Williams, S., Trewartha, G., Cross, M. J., Kemp, S. P., & Stokes, K. A. (2017a). Monitoring what matters: A systematic process for selecting training-load measures. *International Journal of Sports Physiology and Performance*, *12*(s2), S2101–106. https://doi.org/10.1123/ijspp.2016-0337

Williams, S., West, S., Cross, M. J., & Stokes, K. A. (2017b). Better way to determine the acute: Chronic workload ratio? *British Journal of Sports Medicine*, *51*(3), 209–210. https://doi.org/10.1136/bjsports-2016-096589

Windt, J., Ardern, C. L., Gabbett, T. J., Khan, K. M., Cook, C. E., Sporer, B. C., & Zumbo, B. D. (2018). Getting the most out of intensive longitudinal data: A methodological review of workload-injury studies. *BMJ Open*, *8*(10), e022626. https://doi.org/10.1136/bmjopen-2018-022626

Zeileis, A., & Grothendieck, G. (2005). zoo :S3 infrastructure for regular and irregular time series. *Journal of Statistical Software*, *46*(6), 1–27. https://doi.org/10.18637/jss.v014.i06